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The Effect of Salt, Variable Heat Flow, and Increased Carbonate Reef Overburden on Hydrocarbon Generation and Geothermal Potential of Rift Basins

H. H. Banaja, Saudi Aramco, Dhahran, Saudi Arabia

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Rift basins worldwide are highly prospective and contain significant yet-to-find hydrocarbon resources, as well as substantial geothermal potential. The presence and effectiveness of a source rock are crucial for hydrocarbon occurrences in any basin, as it provides the organic matter that can be converted into oil and gas. Similarly, the thermal history and heat flow of a rift basin play a critical role in shaping its geothermal resources, as they influence the temperature and permeability of the rocks, which are essential for the circulation of hot fluids. Organic-rich layers are deposited at different stages of the rift development and are commonly vertically isolated from one prospective area to another by impermeable layers. The quality of source rocks is usually variable within the same stratigraphy; therefore, it is essential to understand the variation in richness throughout the basin. In order for a source rock to be effective, it must be buried to the right temperature within the hydrocarbon generation window, which is ultimately influenced by the source properties, the burial history, and thermal history of the basin. The thermal history of a basin also has a significant impact on the geothermal potential, as it controls the temperature and heat flow that can be harnessed for power generation. Understanding the interplay between these factors is crucial for the exploration and development of both hydrocarbon and geothermal resources in rift basins.

This study aims to investigate the effects of salt and the variation in heat flow on the hydrocarbon generation potential and geothermal resources in a rift basin. By modeling the response of a rift basin to changing basal heat flow, present-day salt geometries, and initial salt deposition thickness, we can gain insights into the complex interactions between these factors and their impact on the hydrocarbon and geothermal potential of the basin. The results of this study will have implications for the exploration and development of hydrocarbon and geothermal resources in rift basins, and will provide a better understanding of the factors that control the thermal history and heat flow of these systems.

Introduction

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Methodology

This study employs a petroleum system modelling software to generate a coarse grid model with 11 layers, constant source rock properties, and unified boundary conditions as a base model. The base model was then altered to test several scenarios to account for the effect of differential heat flow, and variations in present day and initial salt thicknesses on the generation potential of the basin (Figure 1). The scenarios were then blind tested by the data obtained from wells in the area to check the validity of the models.

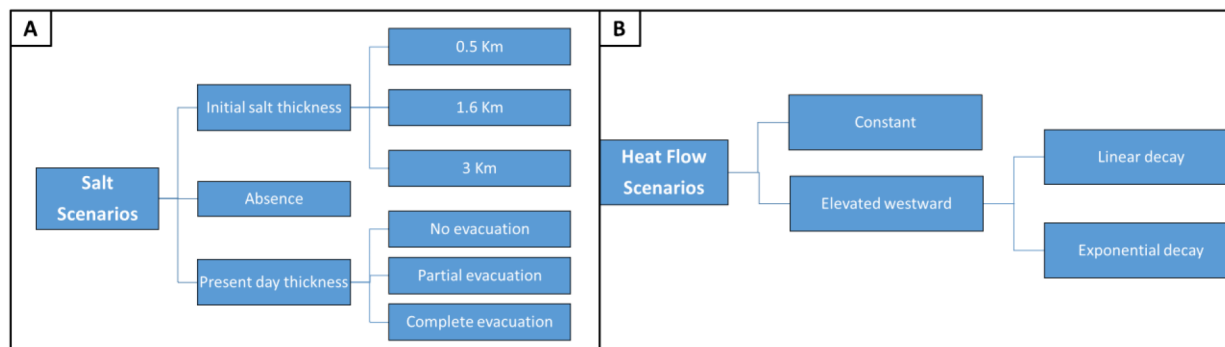


Figure 1—Summary of scenarios modelled with a total of seven scenarios testing the effect of initial and present-day salt geometries (A) and three scenarios testing the variation in heat flow (B).

The base model horizons were gridded to 300 nodes and the cells were extended between horizons. Facies were defined and assigned to each layer based on the regional stratigraphic column. With regards to the boundary conditions, the paleo water depths were inferred from the literature published on each formation and the shape of the basin from the structural reconstruction, while the heat flow was generated using a McKenzie model with a unified stretching factor of 3 and a rifting duration of 35 Ma across the line. The base model assumes 1.6 km of initial salt thickness throughout the basin deposited in a span of 500 thousand years after 11.6 Ma. The salt evacuated the basin gradually from the time of deposition until present-day to form two main salt bodies.

As previously mentioned, the present-day salt geometry shows two salt diapirs in the center of the basin. Several scenarios were modelled to test the effect of present-day salt geometry on the generation potential

of the petroleum system. In the scenario presented, the salt bodies in the east and west have been completely evacuated, while a new salt body has been introduced in the center of the basin. In all the models, the evacuation of salt is assumed to begin immediately after the beginning of the deposition of Layer 3 and continued until the end of deposition of Layer 2. Finally, all these scenarios assume constant basal heat flow and an initial salt thickness of 1.6 km.

In the base model, heat flow was held constant across the basin starting at 50 mW/m² at 65 Ma, rising to a maximum of 70 mW/m² at 20 Ma, finally decreasing to 60 mW/m² at present. To account for differential stretching, heat flow distribution was altered to increase exponentially towards the rift axis while the present-day salt geometry and initial salt thickness were held constant. As a result, heat flow in the eastern part of the model was unchanged from the base model, however, heat flow in the western region of the model was increased from 50 mW/m² at 65 Ma to a maximum of 200 mW/m² at 20 Ma, followed by a decline to 120 mW/m² at present.

Results

The results of this study demonstrate the significant impact of thermal history on the hydrocarbon generation and geothermal potential of rift basins, highlighting the importance of considering basal heat flow and thermal properties of overburden salt in both petroleum and geothermal system modeling. The modeling experiments conducted in this study show that variations in heat flow distribution and salt geometries can substantially influence the source rock "kitchen" and hydrocarbon generation potential, as well as the temperature and heat flow regimes that control geothermal activity, in a rift basin. The results indicate that the thermal history of a rift basin is controlled by a complex interplay of factors, including the basal heat flow, thermal properties of the sediments, and the presence of igneous intrusions and fluids, which in turn affect both the maturation of source rocks and the circulation of hot fluids that drive geothermal systems. The study also highlights the significance of burial history in shaping both the petroleum and geothermal systems, where differences in overburden thickness and erosion can lead to substantial variations in heat flow, hydrocarbon potential, and geothermal potential. The results of this study demonstrate that the thermal properties of overburden salt play a critical role in controlling the thermal history of a rift basin, and that variations in salt geometry and thickness can substantially impact both the hydrocarbon generation potential and geothermal potential of a basin. The modeling experiments conducted in this study also show that the initial salt deposition thickness and present-day salt geometries can have a significant impact on the thermal history, hydrocarbon generation potential, and geothermal potential of a rift basin, highlighting the importance of considering these factors in both petroleum and geothermal system modeling. Overall, the results of this study provide new insights into the complex relationships between thermal history, burial history, hydrocarbon generation potential, and geothermal potential in rift basins, and demonstrate the importance of considering these factors in order to accurately predict the potential for both hydrocarbon and geothermal resources in these systems. The findings of this research could have significant implications for the exploration and production of both hydrocarbons and geothermal energy in rift basins. By considering the complex interplay of factors that control the thermal history of a rift basin, explorationists and production geologists can improve the accuracy of predictions for both hydrocarbon and geothermal resources, and optimize the exploration and production of these resources in a more efficient manner. Results from the base model and subsequently altered models are summarized in [Figure 2](#).

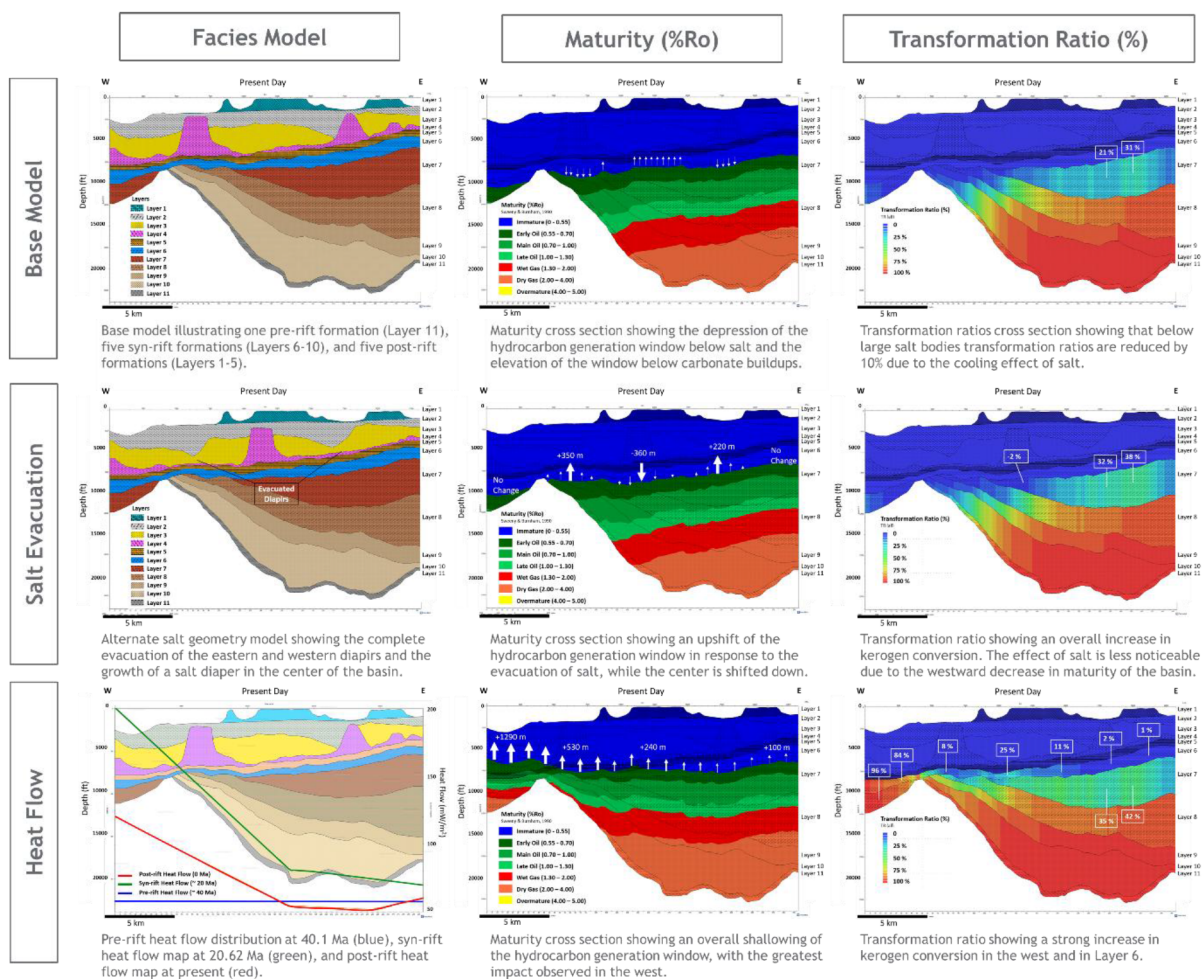


Figure 2—Summary of representative scenarios showing the reduction in maturity below salt, and the increase in maturity at high basal heat flow. It is evident that in the absence of high heat flow (the east), the presence of salt is the main influencing parameter on maturity, while high heat flow will trump the effect of salt on the maturity of the basin and assume the role of most influential parameter.

Effect of salt on hydrocarbon generation potential and geothermal resources

It is evident from the models that the role of salt in the generation potential of a petroleum system is focused on the temperature anomalies created by the presence of salt. The cause of these anomalies is attributed to the contrast in thermal conductivity between the salt bodies and the surrounding sediments, refracting the heat flow towards the center of the diapir robbing heat from the basin vertically, and laterally to a lesser extent. In addition, the geometry and size of the salt bodies has a significant impact on the shape and depth of the temperature anomaly (Mello et al., 1995). In the model presented, the salt layer was first deposited at 11.6 Ma as a massive sheet, where the temperature pattern was exhibited as a steep geothermal gradient compared to the sediments above and below. As the salt moved and accumulated into the eastern and western diapir at present, the observed anomaly was squeezed into an asymmetrical dipolar field centered on the axis of the salt diapirs, increasing the temperature above the salt by 4 °C and decreasing the temperature below by 8 °C. While the values differ, the ratios of the temperature anomalies are consistent with the work of Jensen (1983), who showed that the anomaly above the salt was +10 °C and below was –20 °C. The presented model is further supported by the work of Mello et al (1995) and Hai et al (2016), who illustrate that the depth of the temperature anomaly below the salt bodies is 2 to 3 times the radius of the salt body. On the contrary, the model presented shows that the depth of influence is equal to the radius of the salt

diapir (1 km). The discrepancy in the results could be attributed to several factors. The first explanation is that the volume of the salt body in the literature is just over double that of the model presented. This is important since the models presented show that a 34% reduction in salt volume decreased the depth of influence by half. In addition, the geometries of the salt bodies are different as the diapir in the literature has a height to width ratio of 3:5, while the eastern diapir in the model has a ratio of 1:2. Finally, the contrast between the thermal conductivity of salt and the surrounding sediments in the literature is unknown, and is important because the depth of the temperature anomaly is directly related to the contrast of thermal conductivities. Ultimately, the depth of the cooling effect of the salt will have an impact on the geothermal and generation potential of the petroleum system. Since Layers 8-11 are more than 1 km below the base of salt and more than 90% of the kerogen was transformed prior to the deposition of salt, their present hydrocarbon generation potential will not be affected, however, Layers 5-7 are within the zone of influence. In addition to the thermal anomaly created, the presence of salt will also delay the evolution of the source rock. It was found that increasing the salt thickness from 0 to 2 km, the thermal evolution was delayed by 9 Ma (Hai et al., 2016). The delay effect is validated by the presented model as the source rock evolution in Layer 7 was delayed by 2 Ma as opposed to 9 Ma. The difference in delays could be a result of different sedimentation rates; the rate of heating the deposited sediments is slower than the sedimentation rate (Mello et al., 1995). Furthermore, the Kuqa basin, studied by Hai et al. (2016), reaches a maximum temperature of 200 °C in contrast to 232 °C in the area of interest, which will lower the rate of kerogen conversion (Waples, 1980). In contribution to the present-day salt geometries, the initial salt thickness will also play a role in the rate of transformation of kerogen to hydrocarbon. The presented models show that increasing the initial salt thickness from 0.5 km to 3 km increased the rate of transformation from 0.76% per Ma to 3.79% per Ma.

Effect of heat flow on hydrocarbon generation potential

The base model presented assumes constant heat flow across the basin, however, it is more likely that the heat flow will increase toward the rift axis (Gibbs, 1984; Jarvis and McKenzie, 1980; Abdunaser and McCaffrey, 2014; Fossen, 2018). Plotting the distribution of heat flow across the basin, it is apparent that the western portion of the basin has a higher stretching factor as illustrated in Figure 3. Due to the scale of the basin compared to the scale of the rift system, a linear increase in heat flow towards the rift axis was assumed to be acceptable, however, well calibrations prove that an exponential increase towards the rift axis is more likely. Increasing the heat flow in the basin westward will impact the generation potential of hydrocarbons and the development of geothermal resources by two main mechanisms. First, since the kinetic reactions responsible for the generation of hydrocarbon are temperature dependent (Pepper and Corvi, 1995; Waples, 1980), increasing the heat flow will have a direct positive impact on the maturity by increasing the overall temperature and altering the temperature distribution within the basin. This, in turn, can enhance the geothermal gradient, making the basin more suitable for geothermal power generation. In addition, the conductivity of different lithologies are expected to decrease with increased temperature (Robertson, 1988; Urquhart and Bauer, 2015; Mello et al., 1995). Most notably, salt, sandstone, and limestone, which exhibit exponential decrease in thermal conductivity with increasing temperature compared to shale, which displays a minute linear decrease (Figure 4). As the temperature in the western region of the models increases from <100 °C to >200 °C, the thermal conductivity of the western salt diapir is expected to decrease from 5.5 to 4.0 W/m°C, resulting in a reduced depth of cooling effect caused by the salt due to the decreased contrast in thermal conductivities. The reduction in thermal conductivity of the salt body, in combination with the increased heat flow will result in a significant shift in the hydrocarbon generation window as demonstrated by the model, where the source rock kitchen was elevated by 1.2 km in the west. Furthermore, the increased heat flow and temperature can also enhance the geothermal potential of the basin, making it more suitable for geothermal power generation. In contrast, the temperature of the eastern diapir is slightly elevated from ~50 to about 55 °C, therefore, the thermal conductivity of the eastern diapir is not expected to show a

significant decrease. As a result, the contrast between the thermal conductivity of the salt body and the surrounding sediments is unaffected and the depth of the cooling effect remains the same. The implications of these findings are significant for both hydrocarbon and geothermal exploration, as they highlight the importance of considering the spatial variation in heat flow and thermal conductivity when evaluating the potential of a basin.

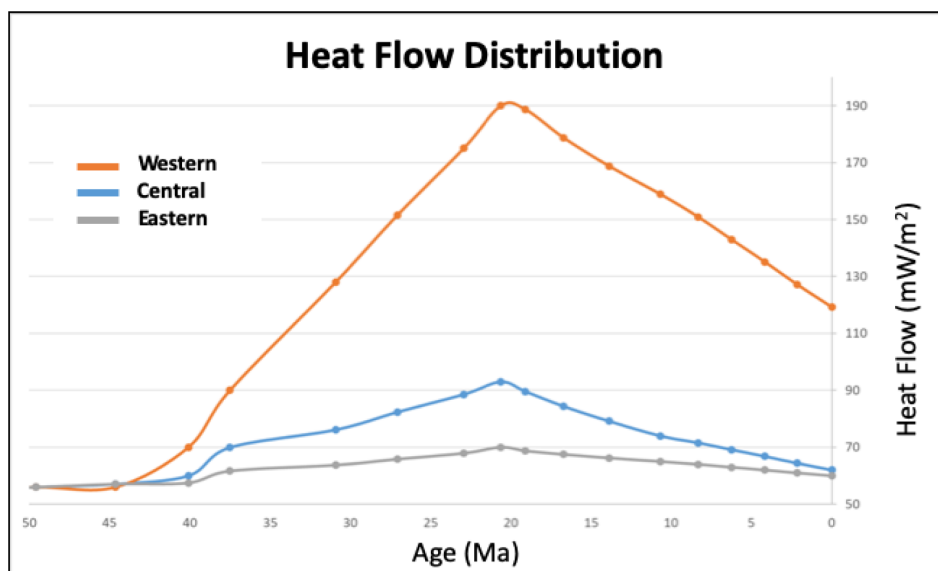


Figure 3—Heat flow trends extracted from the western, central, and eastern parts of the model, showing that the stretching factor in the basin is increasing westward where higher heat flow is modeled.

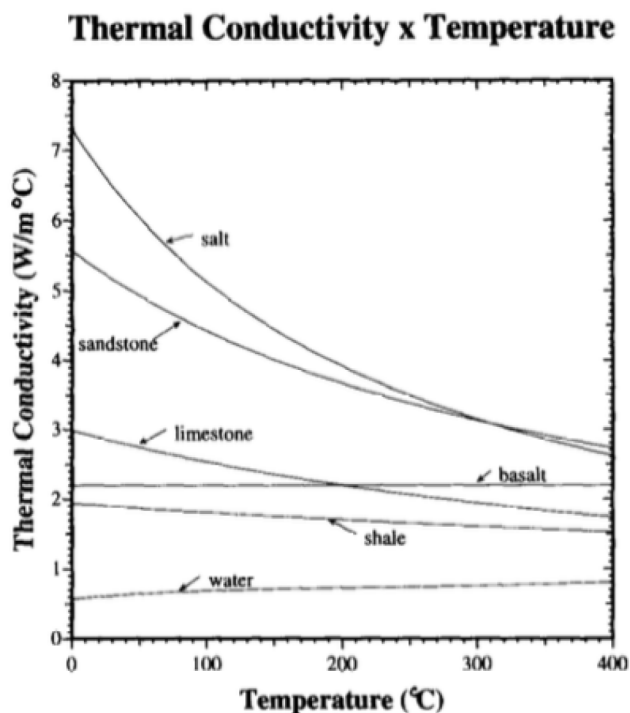


Figure 4—Thermal conductivities of different lithologies as a function of temperature showing an exponential decline in the thermal conductivity of salt and sandstone, compared to the more linear decline in thermal conductivity of limestone and shale (Mello et al., 1995).

Practical implications of the study

The lessons learned from this study can be integrated into various workflows to help identify salt bodies, hydrocarbon prospects, and geothermal prospects. This study shows that heat flow through and around large bodies of salt are significantly different from surrounding sediments due to the contrasts in thermal conductivity. This information is valuable for the identification of salt structures where seismic data is sparse or unavailable and can be obtained from inexpensive temperature data collected from drilling water, exploration, and delineation wells (Jensen, 1983; Poley and Van Steveninck, 1970). In addition, it is also possible to identify the location of salt bodies by studying collapse structures in the overburden reflected on the seabed (Bertoni and Cartwright, 2005), as well as low relief structural highs in the bathymetry of salt basins. The identification of salt bodies can also influence decision making when it comes to hydrocarbon prospects by showing areas where the hydrocarbon generation potential of a basin will be altered from the norm. If not considered, prolific salt basins may show over mature source rocks, where the presence of salt will actually delay the kinetic reactions resulting in a less mature, possibly still producing, source rock (Mello et al., 1995; Hai et al., 2016). Similarly, regions under large salt structures in an early mature basin maybe immature. By studying the temperature data from wells in combination with bathymetry analysis, portions of an over mature basin, for example, may be classified as areas of interest in exploration due to the effects of salt on hydrocarbon generation. Furthermore, the identification of salt bodies can also inform geothermal exploration by highlighting areas with enhanced heat flow and temperature anomalies, which can be exploited for geothermal energy production. The combination of low conductivity surface sediments and enhanced heat flow above bodies of salt create abnormally high temperatures that may be used for power generation, space heating, or other industrial applications (Nagihara et al., 1992; Jensen, 1983). Additionally, the temperature gradients around salt diapirs can also be used to identify potential geothermal reservoirs, where hot water or steam can be extracted for energy production. The understanding of convective flow and its role in distributing quartz cementation within a reservoir (Haszeldine et al., 1984) can also be applied to geothermal systems, where the movement of hot fluids can be critical for maintaining permeability and enhancing heat transfer. Overall, the integration of this study's findings into exploration workflows can help to identify new hydrocarbon and geothermal prospects, and optimize the development of existing ones.

Limitations of the study

It is essential to understand that salt structure, heat flow, and overburden are only a small fraction of the influential parameters dictating hydrocarbon generation and geothermal potential. In order to fully understand how these parameters are interacting and which input is the most sensitive, statistical methods should be applied, similar to the study of Tong and Mukerji (2017). This approach can help to quantify the uncertainties associated with each parameter and identify the most critical factors controlling hydrocarbon generation and geothermal potential. Additionally, the structural reconstruction in the model was performed by the restoration of the master fault and guided by stratigraphic projections, which in a syn-rift section, can lead to great uncertainties. For a more reliable structural reconstruction, additional methods such as uplift from vitrinite reflectance projection and uplift from porosity-dependent variables should be employed to complement the restoration (Byrnes and Lawyer, 1999). Similarly, the halokinetic restoration presented in the models is one variation of many possible scenarios. The timing of the salt movement and geological development of the basin is crucial to accurately estimate the thermal history of the basin, which in turn affects both hydrocarbon generation and geothermal potential (Mello et al., 1995). Furthermore, the study assumes that the salt diapirs are uniformly composed of a specific mixture of halite and anhydrite, whereas large salt structures tend to display a complex variability in intrasalt structural style and composition, as well as the possibility of allochthonous inclusions within the salt (Jackson et al., 2015). This complexity can have significant implications for both hydrocarbon and geothermal systems, as the thermal conductivity and rheology of the salt can vary significantly depending on its composition and structure. Therefore,

future studies should aim to incorporate more detailed and realistic representations of salt composition and structure, as well as other influential parameters, to improve our understanding of hydrocarbon generation and geothermal potential in salt-bearing basins.

Conclusions

The sensitivity of different parameters in any basin model is highly contextualized due to the complex interaction of boundary conditions, input parameters, and the distribution of sedimentological facies within the basin. For the purpose of this study, the models were designed to test the impact of present-day salt geometry, initial salt deposition thickness, and lateral variations in heat flow on hydrocarbon generation potential and geothermal resources. The results from the scenarios modeled show that the most impactful input parameter varies spatially. It was found that the hydrocarbon generation potential in the eastern part of the model is mostly influenced by the presence of salt, while heat flow appears to be the main variable influencing hydrocarbon generation in the west. Similarly, the geothermal potential of the basin is also affected by the interaction of salt and heat flow, with areas of high heat flow and salt-induced thermal anomalies showing promise for geothermal energy production. Several assumptions can be made to address the causes of this variation in sensitivity. As previously discussed, the negative exponential relationship between heat and thermal conductivity of salt suggests that as heat flow increases, the effect of salt on the hydrocarbon generation potential of a basin will decrease, as is the case in the western part of the model (Robertson, 1988; Urquhart and Bauer, 2015). On the contrary, the influence of salt will become more apparent in low heat flow regions, such as the eastern part of the model, where the salt-induced thermal anomalies can have a significant impact on hydrocarbon generation and geothermal potential. Furthermore, the results of this study highlight the importance of considering the spatial variability of heat flow and salt distribution when evaluating the hydrocarbon and geothermal potential of a basin. By integrating these factors into basin models, explorers and researchers can better understand the complex interactions that control hydrocarbon generation and geothermal resources, and make more informed decisions about exploration and development strategies.

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